Repeating Efficiently

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Test before repeating. Keep the stopping evidence.

Trace current-state loops, compare zero iterations and overshoot, diagnose failed progress and try a countdown.

Keep one habit: Test the current state → execute the body if authorised → return to the test.

MAIN LESSON · SECTION 1

When does the next repetition happen?

> Test before each body execution, count iterations separately and keep the actual final value.

Start with a stopping test

A loop can test four times and update only three times. Its last false test explains why it stops. A value seen during a run is not necessarily a final report.

Retrieve the one-decision rule

Score is 4. If score is below 5, add 3; otherwise subtract 2. Report and stop. Does reaching 7 cause another test? Explain.

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can trace a **loop** by testing its condition before every repetition.
- I can distinguish the number of repetitions from the final value and include zero repetitions.
- I can explain why an update reaches a **stopping condition** or why a faulty **loop** keeps repeating.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

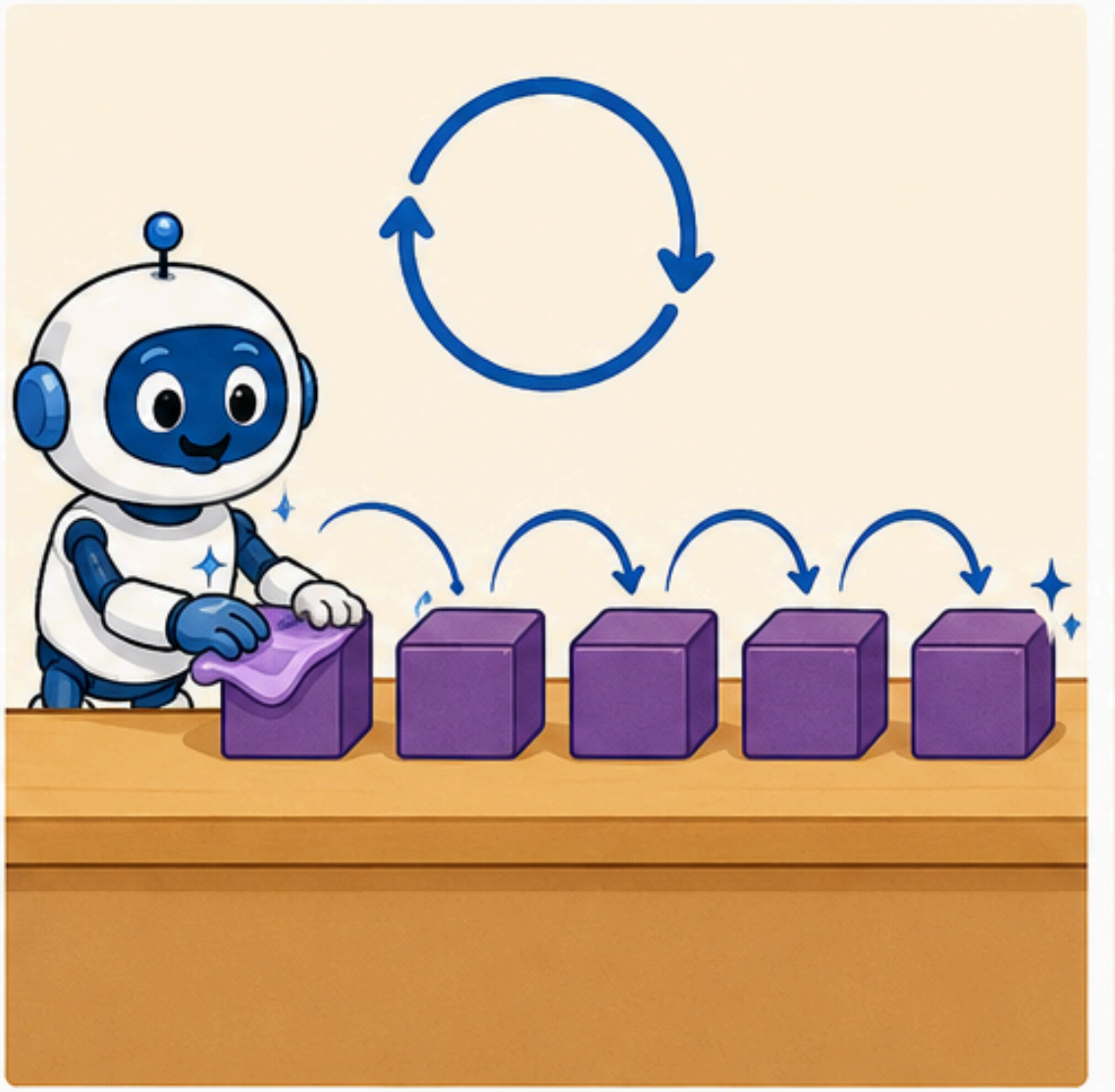
Remember before reading

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

A **loop** repeats a useful step until a **stopping condition** is met.



LEARNING TARGET A robot stamps the next unstamped card until it has stamped five cards. Only three cards are available. Write a stopping rule that handles both limits and trace the number stamped.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can check a **loop** condition before each update and count **iteration**s.

MAIN LESSON · SECTION 2

A while loop returns to its test

> Use the updated counter each time the body finishes.

Read the below-7 loop

A paper counter starts at 1. While the counter is below 7, add 2 to it. When the test is false, report the counter and stop.

The word while means test first, run the **body** only if true, then return to the test using the updated value. The **body** here is just add 2.

Count tests and additions separately

The counter follows 1, 3, 5, 7. At 1, 3 and 5 the condition is true, so each test is followed by one addition. At 7 the condition is false. No addition follows that final test. The **loop** has three **iteration**s and reports 7.

Contrast while with one if decision

Compare the Week 10 instruction if counter is below 7, add 2, then report. From 1 that single decision reports 3. A while **loop** explicitly returns to its test; an if decision does not.

Draw a pointer to the instruction being executed instead of deciding for yourself when to repeat.

Explain the stopping step

- For the start at 1, mark each true test and the addition it authorises. What happens at the first false test? Explain why a single if decision from the same start behaves differently.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY Instead of writing draw-side and turn four times, a square algorithm can repeat those steps four times.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Keep the final false test in the trace

> Each row needs the current value, condition result and authorised next action.

Read the complete test sequence

A complete loop trace

COUNTER BEFORE TEST	IS COUNTER BELOW 7	NEXT ACTION
1	True	Add 2 to reach 3
3	True	Add 2 to reach 5
5	True	Add 2 to reach 7
7	False	Report 7 and stop

Read the rows from top to bottom as one run. Each true test performs one addition and returns to the test. The final false test performs no addition.

Explain four tests but three iterations

The trace has four tests but only three **iteration**s. The false test is **evidence** that this run stops. A list ending at 5 would be incomplete, while a list continuing to 9 would include an addition the **loop** did not authorise.

Retain repeated states as evidence

Write the current value, the test result and the action on every row. Repeated values can matter: if the **body** does not change the state used by the test, the same true test may occur again.

Do not erase these rows simply because they look unhelpful.

Explain a deciding row

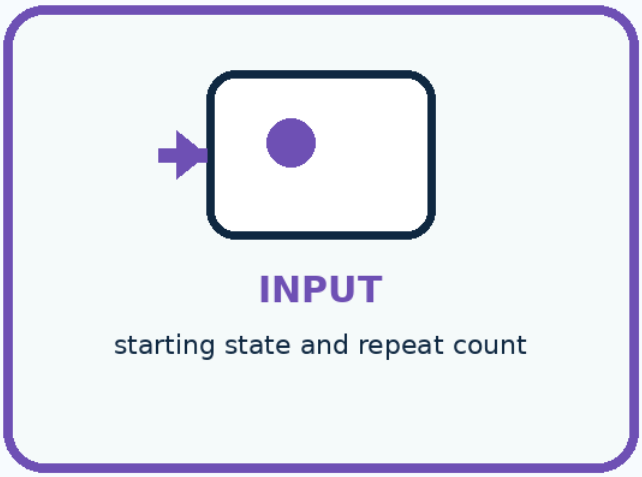
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

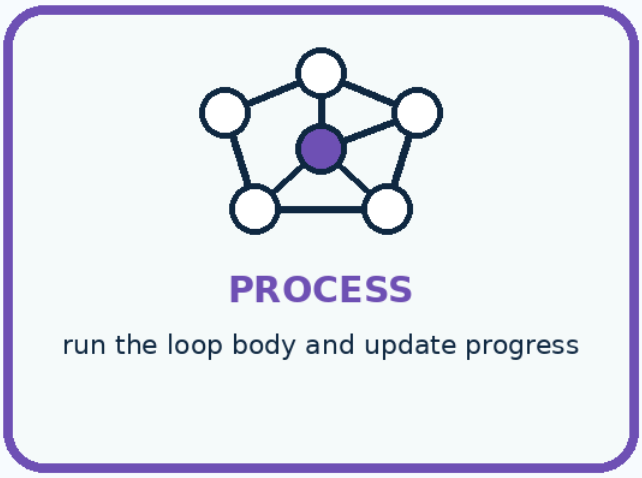
▼ Earlier wording, alternatives and picture transcripts

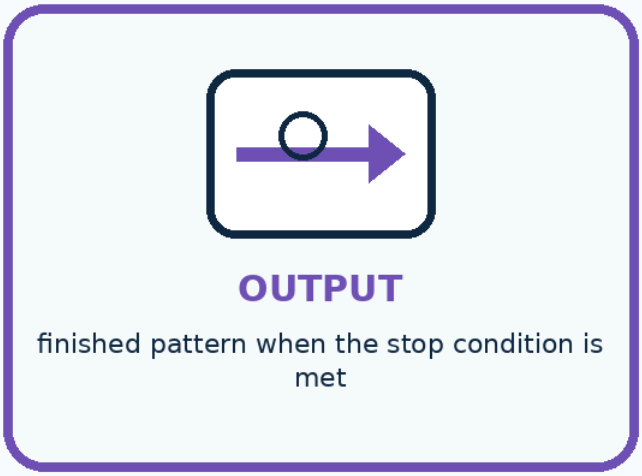
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT starting state and repeat count

PROCESS run the **loop body** and update **progress**

OUTPUT finished pattern when the stop condition is met

CORE IDEA A **loop** repeats a useful step until a **stopping condition** is met.

UNDER THE HOOD For these bounded counter exercises, specify the starting value, repeated action, update and stopping test. Check that the updates eventually make the test false; otherwise a run may continue forever or repeat the wrong number of times. Other programs may use different **loop** designs, such as waiting for a stop signal.

Draw a fresh example with all three stages:

A closer explanation

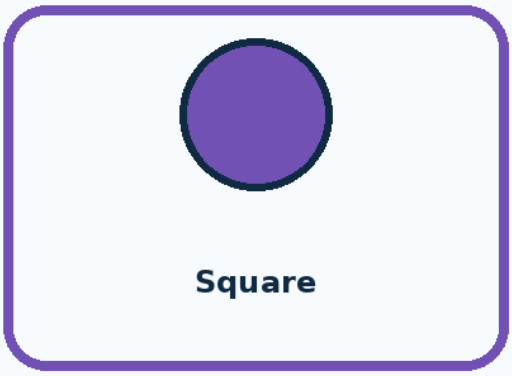
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare repetition and stopping clues

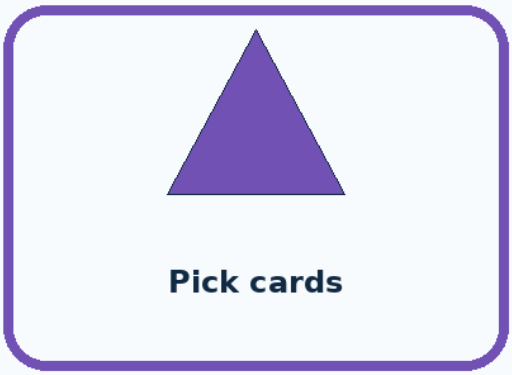
> Identify the body, the next test and what makes repetition end.

Pictures paired with their evidence



The case to investigate

CASE 1 — Square Clue: repeat move and turn 4 times. Meaning: counted **loop**.



The case to investigate

CASE 2 — Pick cards Clue: repeat until pile empty. Meaning: condition **loop**.



The case to investigate

CASE 3 — No counter update Clue: condition never changes. Meaning: infinite **loop** risk.



The case to investigate

CASE 4 — Stop too early Clue: one item remains. Meaning: off-by-one error.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

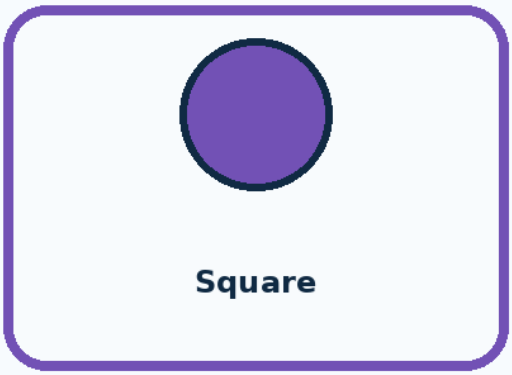
BOUNDARY Repeat forever is not an efficient algorithm unless endless operation is truly the goal and can be stopped safely.

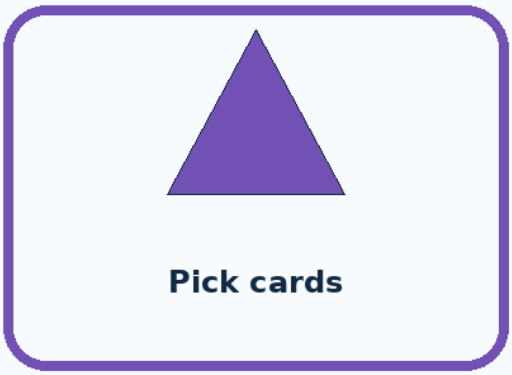
▼ **Earlier wording, alternatives and picture transcripts**

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Square Clue: repeat move and turn 4 times. Meaning: counted **loop**.

CASE 2 — Pick cards Clue: repeat until pile empty. Meaning: condition **loop**.

CASE 3 — No counter update Clue: condition never changes. Meaning: infinite **loop** risk.

CASE 4 — Stop too early Clue: one item remains. Meaning: off-by-one error.

Change one thing in the pictured case

Change: Start at 10 instead of 2.

Prediction: The **loop** runs zero times and reports 10.

Reason: Its first condition is already false, so no addition happens.

The loop may cross the boundary before stopping

> Use the stated add-3 loop and report the actual state after its false test.

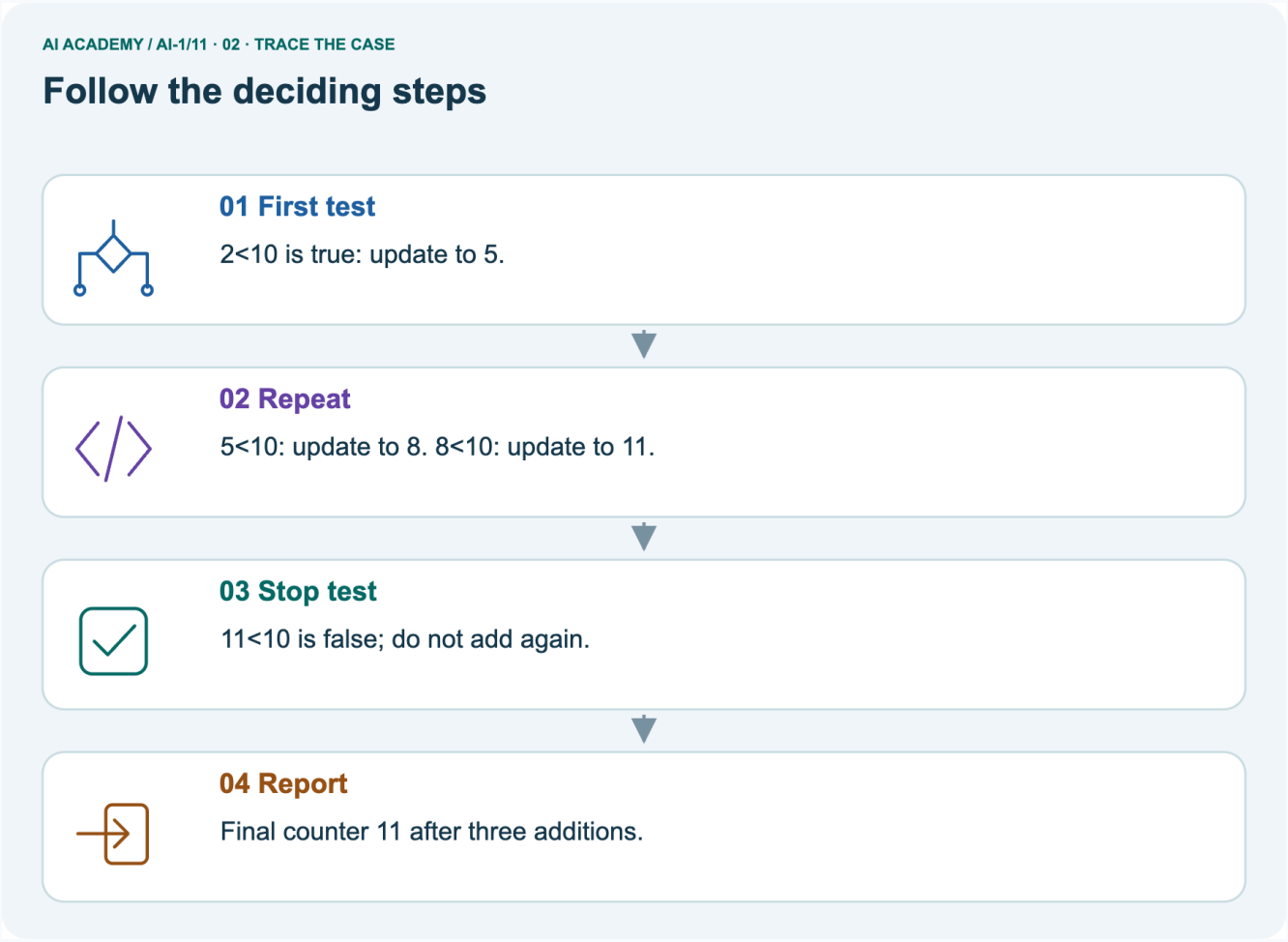
The new counter recipe

A fictional paper counter advances through a display until its condition becomes false.

Start at 2. While the current counter is below 10, add 3.

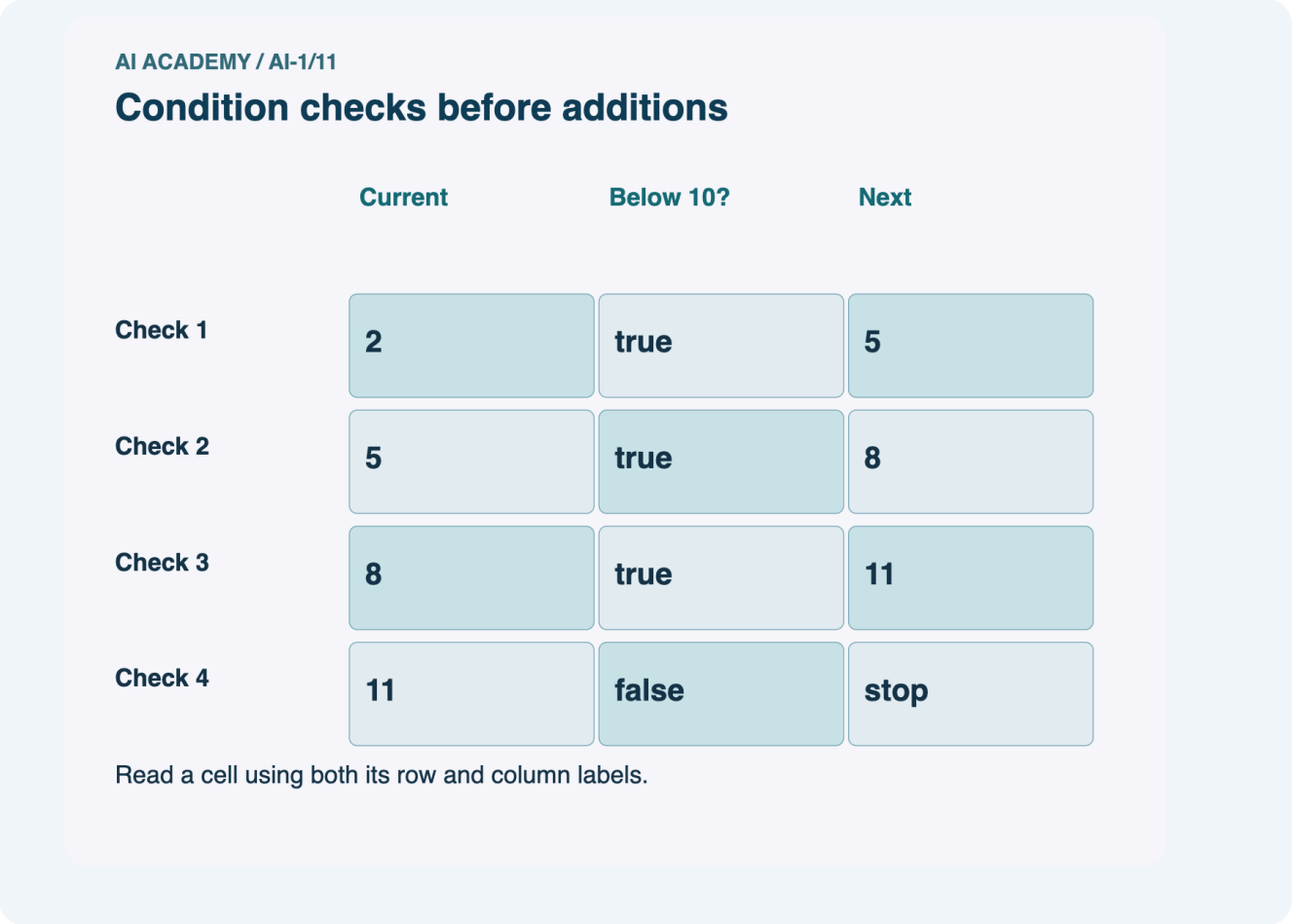
Test before each addition; report when the condition is false.

Follow every test and update



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare the condition-check record



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why 11 is a valid report

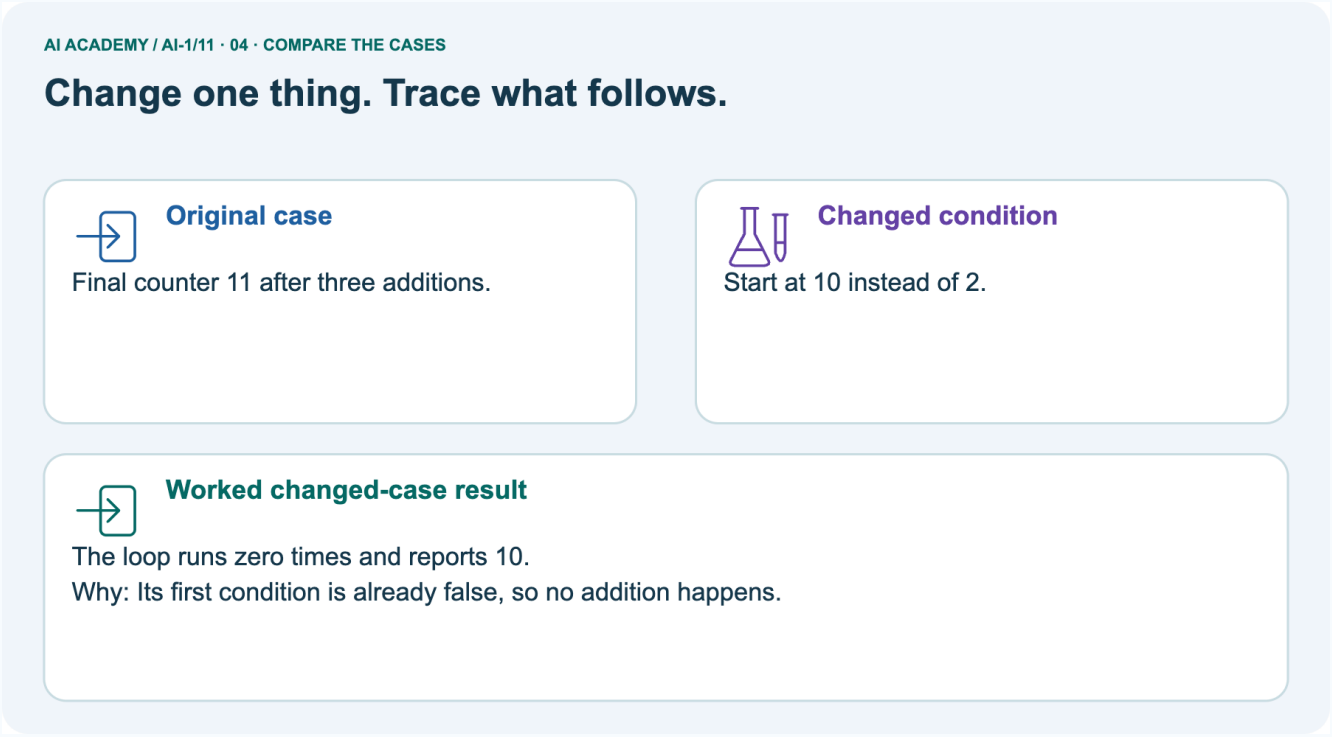
The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10. The final 11 is therefore allowed by this procedure.

Record both the condition and counter to distinguish a correct stop from an assumed endpoint.

Predict from the boundary itself

- Before checking, start this below-10, add-3 loop at 10. Write its first test and decide whether any addition is authorised. Keep the iteration count separate from the reported value.

▼ Check the zero-iteration run



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the need for progress

This **loop** stops because positive increments eventually cross the limit. Other updates can fail to make **progress**.

Trace the fresh start at 9

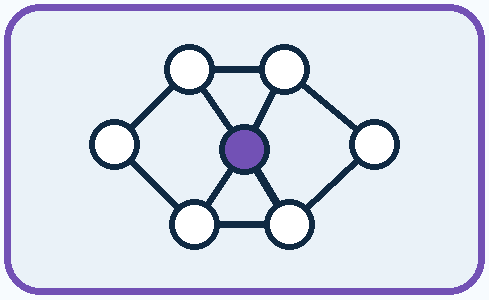
- Read the labelled picture: Trace a fresh start of 9. How many additions occur, and must the report equal 10?

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE Instead of writing draw-side and turn four times, a square algorithm can repeat those steps four times.

STEP 1 — OBSERVE starting state and repeat count

STEP 2 — TRACE run the **loop body** and update **progress**

STEP 3 — CONCLUDE finished pattern when the stop condition is met

MODEL REASONING Place only the repeated steps inside the **loop**, initialize the counter, update it once per pass, and stop after the required count. Trace each **iteration** to catch one-too-many or one-too-few errors.

DO NOT OVERCLAIM Repeat forever is not an efficient algorithm unless endless operation is truly the goal and can be stopped safely.

What new **evidence** could change this answer?

A closer explanation

Starting counter	Values at successive tests	Iterations and reported value
5	5 true then 7 false	1 iteration and report 7
6	6 true then 8 false	1 iteration and report 8
7	7 false	0 iterations and report 7
8	8 false	0 iterations and report 8

While tests before the **body**. If the first test is false, the **body** runs zero times. Do not add 2 once just because a **loop** is present. Starting at 7 or 8 is a useful test of this rule.

A **loop** that stops when the counter is no longer below 7 does not promise to finish at exactly 7. Starting at 6 produces 8. Passing the boundary is allowed by these instructions.

Always report the actual final state instead of replacing it with the threshold.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Repeating Efficiently

The facts we will use

AI ACADEMY / AI-1/11

Condition checks before additions

	Current	Below 10?	Next
Check 1	2	true	5
Check 2	5	true	8
Check 3	8	true	11
Check 4	11	false	stop

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/11

Repeating Efficiently

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First test

2<10 is true: update to 5.



2. Repeat

5<10: update to 8. 8<10: update to 11.



3. Stop test

11<10 is false; do not add again.



4. Report

Final counter 11 after three additions.

CHANGE ONE THING

Start at 10 instead of 2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First test	2<10 is true: update to 5.
2. Repeat	5<10: update to 8. 8<10: update to 11.
3. Stop test	11<10 is false; do not add again.
4. Report	Final counter 11 after three additions.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Start at 10 instead of 2. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Progress must fit the stopping test

> Growing numbers do not guarantee that an equality target will be reached.

Why the even-number run never reaches 7

A different mistake says while counter is not equal to 7, add 2. From 6 the values are 6, 8, 10 and then larger even numbers. None is 7, so this equality test never becomes false.

A growing number does not guarantee termination: the update must be able to reach a state that makes this particular test false.

Preserve an unavailable starting value

Our paper examples use known whole numbers without a maximum size. For this lesson, an unreadable starting counter means pause and request the value. It does not mean start at zero.

Later programs will need explicit rules for unavailable **input**s and resource limits.

Choose revealing additional starts

- CLAIM A A **loop** is just copying the same words many times.
- CLAIM B The **stopping condition** will become true by itself.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **A tester runs the below 7 loop only from starting counter 1. Choose one additional start that checks zero repetitions and one that checks passing the boundary without landing on it. Give the expected reported values**

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Square	repeat move and turn 4 times	counted loop
Pick cards	repeat until pile empty	condition loop
No counter update	condition never changes	infinite loop risk
Stop too early	one item remains	off-by-one error

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: distinguish a stopped trace from an interrupted one

> Show why the update eventually makes the test false, or why it keeps returning the same result.

Explain progress for positive increments

For our known whole-number counters below 7, adding 2 makes the counter larger each time. After enough additions it is at least 7, so the test becomes false.

This explains why this **loop** stops for these **input**s; one successful trace alone would not establish that for every possible procedure.

Trace the faulty add-0 body

Now change the **body** to add 0. Starting at 1 gives the same value 1 at every test. Below 7 remains true, and the **loop** never reaches its reporting instruction.

A classroom trace should be stopped by the teacher after the repeated state is established. That intervention is not a successful program result.

> **Compare the original add 2 loop from starting counter 3 with the faulty add 0 version. Record enough test rows to justify each conclusion. Explain the difference between the final reported value and a value merely seen during a non stopping run**

Use W2 to check the very first test

Use workbook W2 for the connected example “**Loop**s and stopping rules”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “every loop runs at least once”

- Use the worked case for **Loop**s and stopping rules. Proposed explanation: “Every **loop** runs at least once”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured overshoot

Trace a fresh start of 9. How many additions occur, and must the report equal 10?

My explanation using a pictured step and the given facts

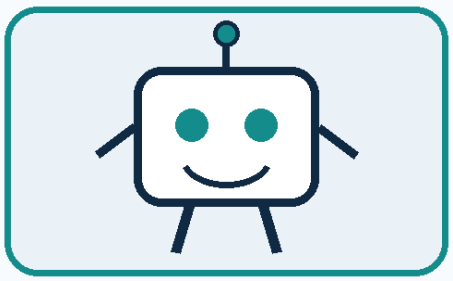
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A **loop** repeats a useful step until a **stopping condition** is met.

Use a fresh example to explain **loop**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: The **stopping condition** will become true by itself.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Check the first test

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate a complete drawing-loop card

> Compare a normal loop, zero repetitions and a missing progress update.

The initial state, body and stopping rule

Use a paper row, a next-cell marker and a counter drawn. Start with drawn=0 and target=3. While drawn is below target: draw a star in the next empty cell, move the marker one cell right, then add 1 to drawn. When the test is false, report drawn and stop. A loop card must include that entire three-instruction body.

Predict, trace and diagnose

1. Predict the picture and trace target=3. Reset and try target=0; record the first false test without drawing. Then inspect a faulty card that omits add 1 to drawn. On paper, show two repeated true tests and explain why the target is never reached in that faulty trace; stop the classroom demonstration there. That intervention is not the procedure successfully reporting a result.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep iterations, tests and reports separate

Case	Prediction	Observed result	What it means
Normal			
Edge			

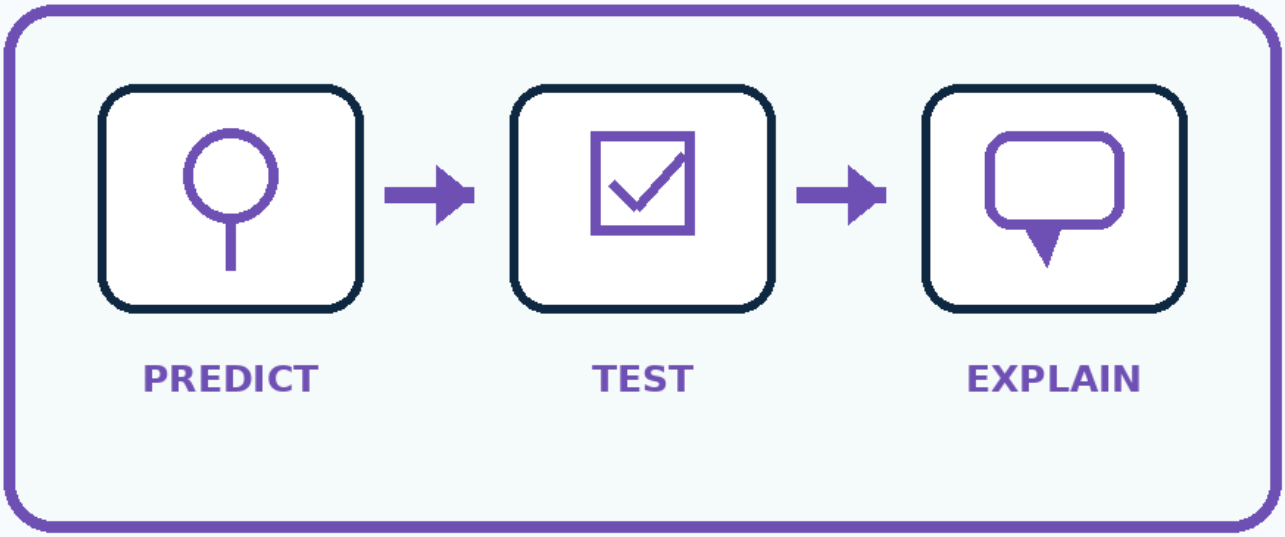
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Loop Card Drawing

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Compress a repeated drawing algorithm using a **loop** card.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: transfer the loop method to a countdown

> Use the above-1 test and subtract-2 body, then inspect zero progress.

The new countdown task

A paper countdown starts at 7. While remaining is above 1, subtract 2. When false, report and stop. Trace the run and count **iteration**s. Repeat from 6 and from 1. Explain why the final value need not equal 1.

What happens from 7 if subtract 2 is replaced by subtract 0?

My independent explanation, trace and limits

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer to a countdown

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Repeat clap then tap three times. Write the complete sequence. How many actions occur? What would happen if the stopping instruction were missing?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Use a **loop** to **process** each item card in the test pile, with an explicit stop when no cards remain.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A robot stamps the next unstamped card until it has stamped five cards. Only three cards are available. Write a stopping rule that handles both limits and trace the number stamped.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Give repeated sorter work a stopping rule

> Check both an empty input pile and a normal finite pile.

This week’s project decision

THIS WEEK'S CHECKPOINT Use a **loop** to **process** each item card in the test pile, with an explicit stop when no cards remain.

Give any repeated sorter step an explicit **stopping condition** and trace a zero-**iteration** example.

Plan the state, updates and tests

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep endless repetition distinguishable from useful stopping

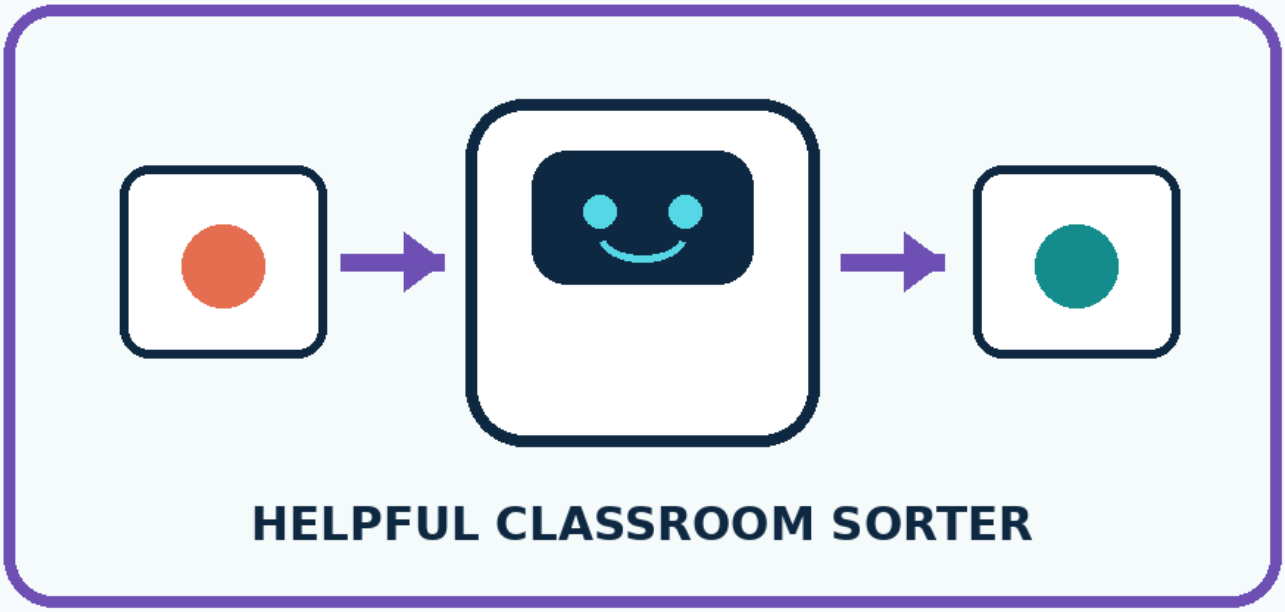
RESPONSIBLE DESIGN Repeat forever is not an efficient algorithm unless endless operation is truly the goal and can be stopped safely.

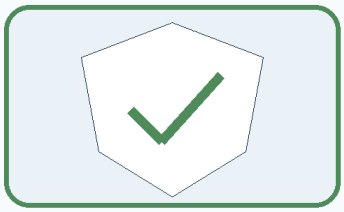
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: justify the final false test

> Close your notes and explain the iteration count and actual report separately.

Your exit explanation

- Close your notes. Explain this goal in your own words: Trace a **loop** by testing its condition before every repetition

> **My deciding step, evidence and one remaining question**

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
loop	control structure that repeats steps	
iteration	one pass through the loop	
loop body	steps repeated each time	
stopping condition	rule that ends repetition	

Terms for the connected explanation

Word	Meaning
Body	The instructions performed on each iteration
Progress	A change that brings the state closer to a stopping condition

Repair a claim using a word

- CLAIM TO REPAIR A **loop** is just copying the same words many times.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ **Earlier wording, alternatives and picture transcripts**

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

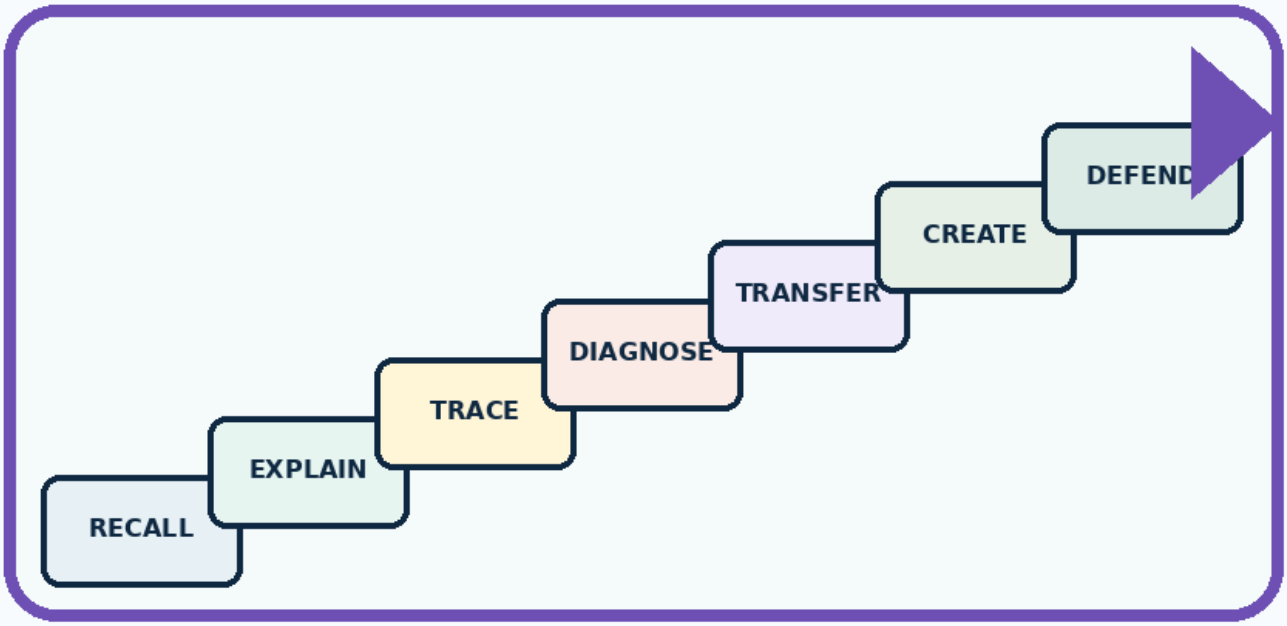
- RECALL Define **loop**.
- EXPLAIN A **loop** repeats a useful step until a **stopping condition** is met.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: A **loop** is just copying the same words many times.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Transfer to a countdown is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

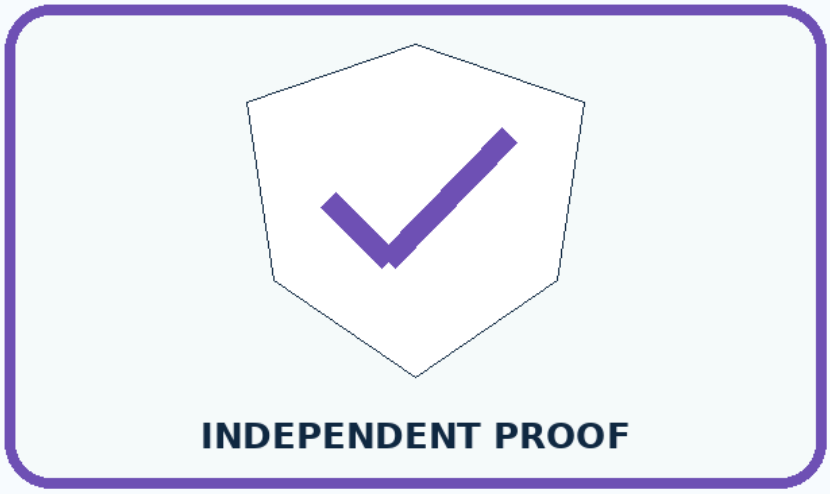
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Transfer to a countdown is the selected independent case.



EXPLANATION CHECK Record every test including the final false one; count body executions, report the actual state and explain progress or nontermination.

Independent **evidence**:

SELF-CHECK Place only the repeated steps inside the **loop**, initialize the counter, update it once per pass, and stop after the required count. Trace each **iteration** to catch one-too-many or one-too-few errors.

NEXT BRIDGE Carry today's **evidence** habit into: Debug the Cause.

▼ **Extra practice: Repeat with a clear stopping point**

REFERENCE · SECTION 13

Extra practice: Repeat with a clear stopping point

> Try the question before opening its worked explanation.

Read the supplied case and question

Repeat take the next unstamped card and stamp it four times.

How many cards are stamped? What assumption matters?

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Four, provided four unstamped cards exist and each repeat takes the next one. A limit or end-of-stack check prevents continuing indefinitely.

Evidence or stage	Value or result
Before	0 stamped
Repeat 1	1 stamped
Repeat 2	2 stamped
Repeat 3	3 stamped
Repeat 4	4 stamped, stop

▼ Earlier wording, alternatives and picture transcripts

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Repeat with a clear stopping point

Repeat take the next unstamped card and stamp it four times.

How many cards are stamped? What assumption matters?

Four, provided four unstamped cards exist and each repeat takes the next one. A limit or end-of-stack check prevents continuing indefinitely.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Trace a loop by testing its condition before every repetition	Explain the deciding step, show a trace or test, and state one limit.
Distinguish the number of repetitions from the final value and include zero repetitions	Explain the deciding step, show a trace or test, and state one limit.
Explain why an update reaches a stopping condition or why a faulty loop keeps repeating	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional paper counter advances through a display until its condition becomes false.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional paper counter advances through a display until its condition becomes false.
- Which labelled step matches this detail? Final counter 11 after three additions.
- 1. Report
- 2. First test
- 3. Repeat
- 4. Stop test
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report Final counter 11 after three additions. The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10.

The final 11 is therefore allowed by this procedure. Record both the condition and counter to distinguish a correct stop from an assumed endpoint.

Changed-case question

- Changed-case practice: Start at 10 instead of 2. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The **loop** runs zero times and reports 10.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The **loop** runs zero times and reports 10. Its first condition is already false, so no addition happens.

Repair and save your work

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which **input** detects the difference between the two written conditions?

Explain its entire run.. Return to this week and explain what you changed.

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

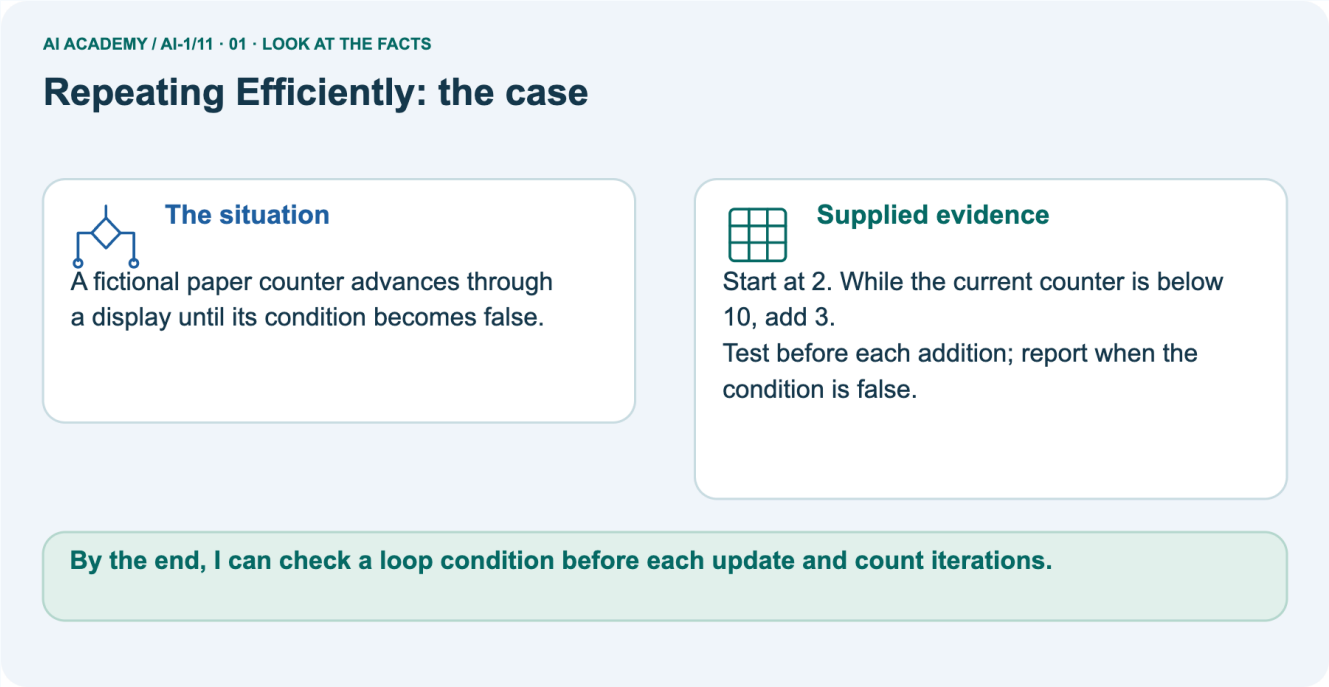
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Repeating Efficiently: the case

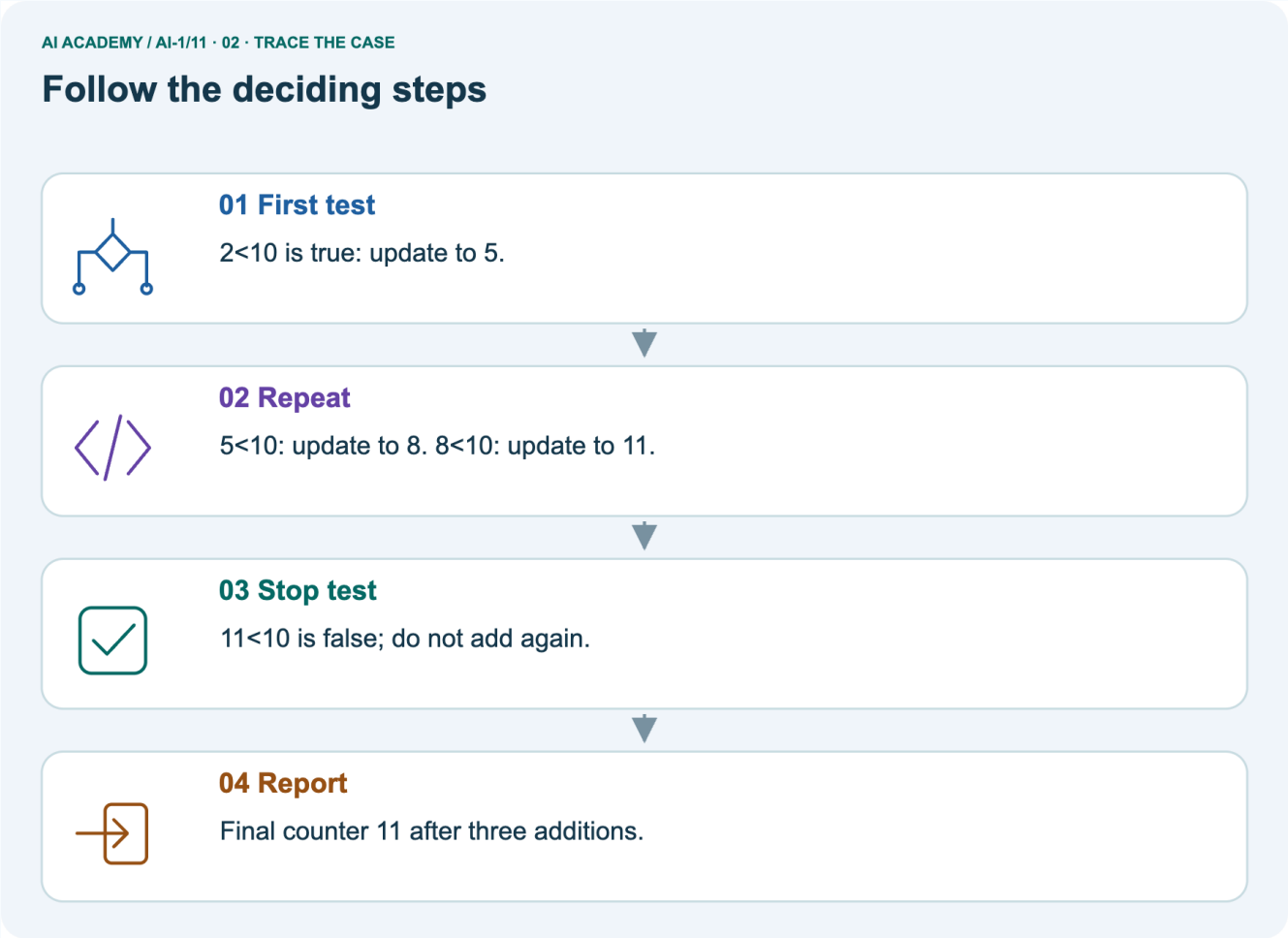


Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper counter advances through a display until its condition becomes false.
Supplied facts: Start at 2. While the current counter is below 10, add 3. Test before each addition; report when the condition is false.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

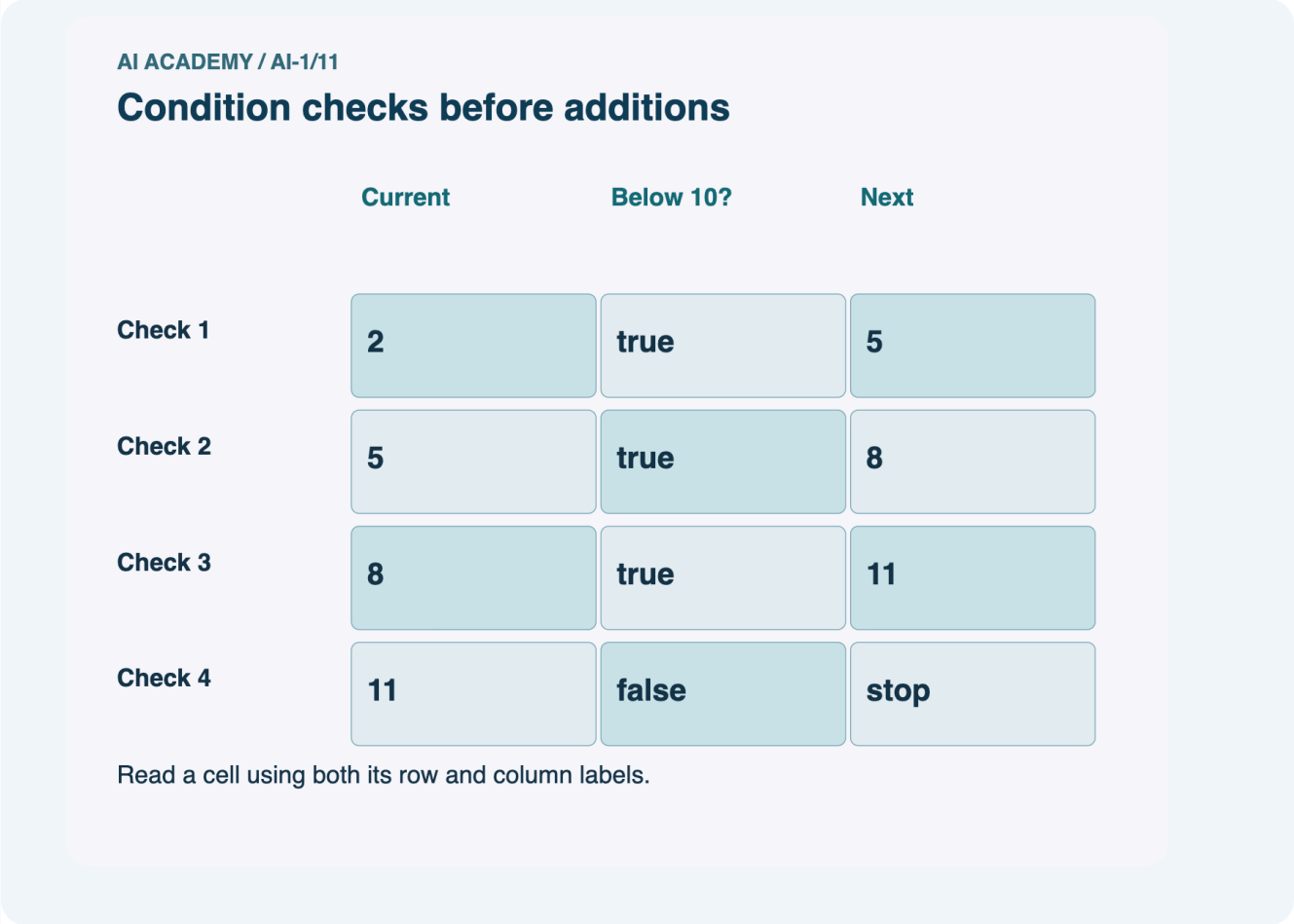


Follow the labelled picture with the matching explanation. Open the vector diagram

First test: 2<10 is true: update to 5. → Repeat: 5<10: update to 8. 8<10: update to 11. → Stop test: 11<10 is false; do not add again. → Report: Final counter 11 after three additions.

Point to each numbered stage and explain which recorded **input** it uses.

03. Condition checks before additions



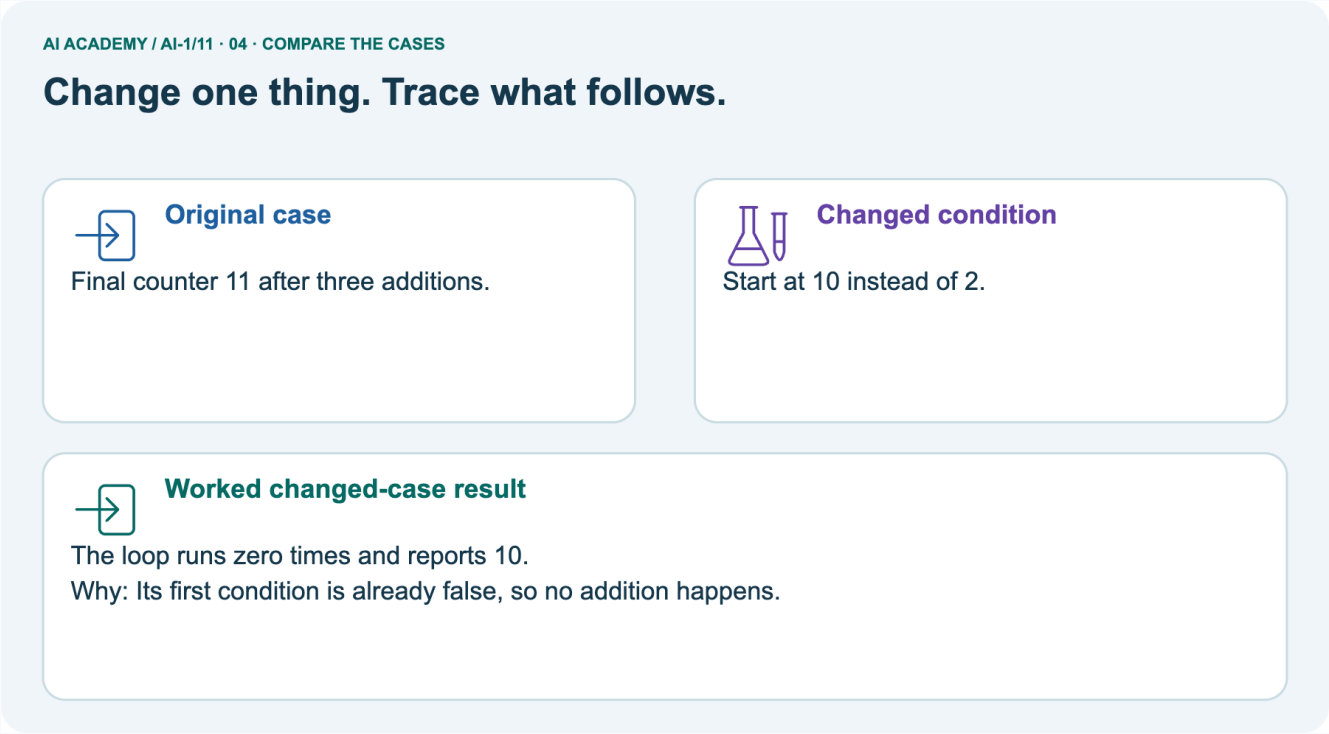
Follow the labelled picture with the matching explanation. Open the vector diagram

Condition checks before additions. Check 1: Current = 2, Below 10? = true, Next = 5. Check 2: Current = 5, Below 10? = true, Next = 8. Check 3: Current = 8, Below 10? = true, Next = 11.

Check 4: Current = 11, Below 10? = false, Next = stop.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



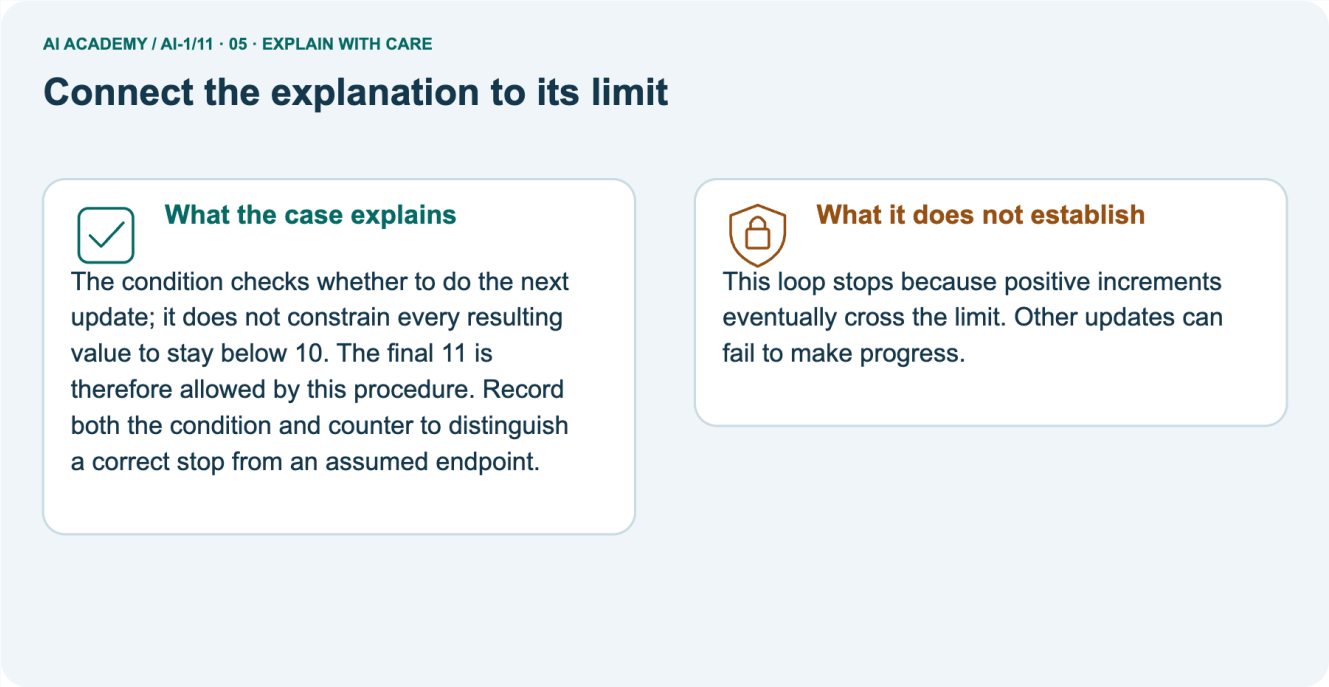
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Final counter 11 after three additions. Changed condition: Start at 10 instead of 2. Expected result: The **loop** runs zero times and reports 10.

Reason: Its first condition is already false, so no addition happens.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



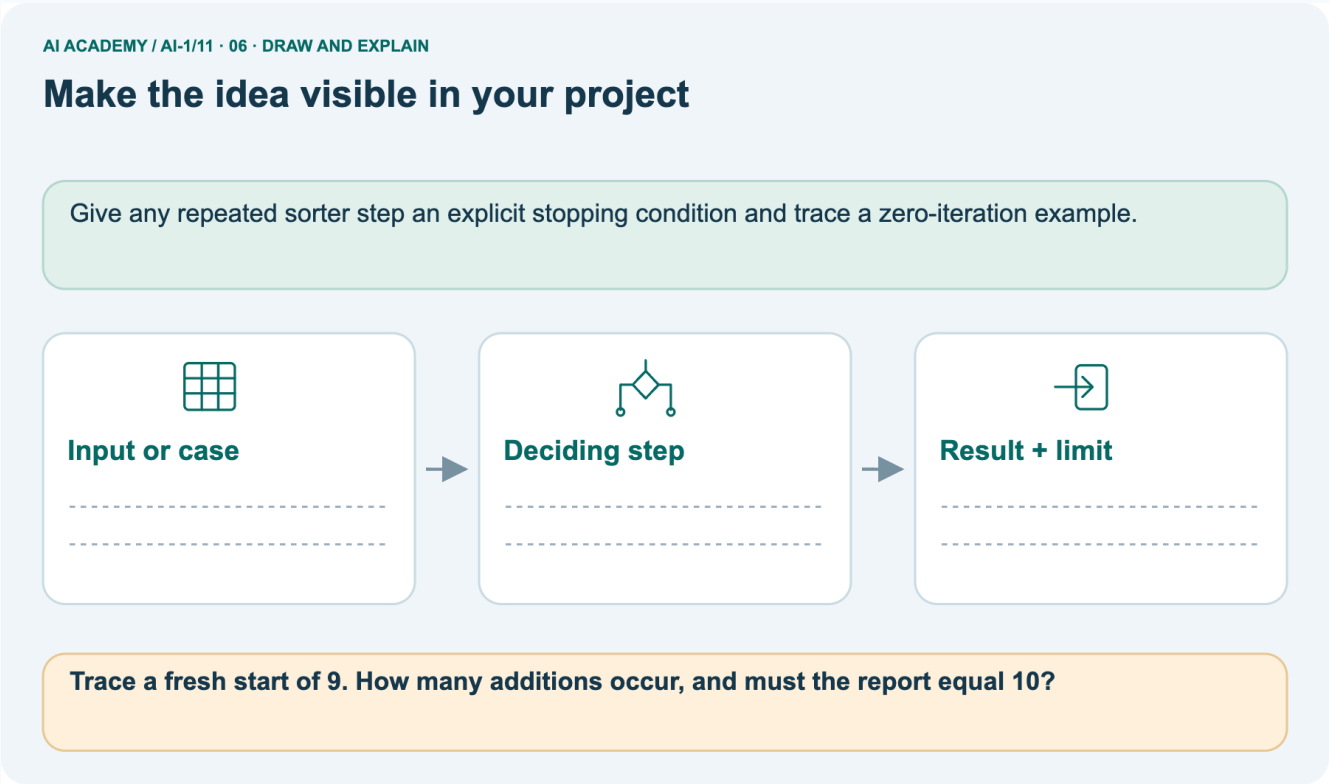
Follow the labelled picture with the matching explanation. Open the vector diagram

The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10. The final 11 is therefore allowed by this procedure.

Record both the condition and counter to distinguish a correct stop from an assumed endpoint. Limit: This **loop** stops because positive increments eventually cross the limit. Other updates can fail to make **progress**.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give any repeated sorter step an explicit **stopping condition** and trace a zero-**iteration** example. Draw your **input**, deciding step, result and limit. Supported question: Trace a fresh start of 9.

How many additions occur, and must the report equal 10?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Repeating Efficiently: the case

02. Follow the deciding steps

03. Condition checks before additions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.